Python Application Developer – Dept of Finance – Hays ACT 17/7/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Python Application Developer/ Integration Solution Architect/Software Engineer/Full Stack Developer(Java/.NET/Python) / Snr DevOps Engineer /Snr Data Engineer/Mulesoft/AI/ML/DL/BI/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Python Application Developer/Software Engineer/Snr DevOps/Data Engineer/AI/ML/DL/SP/D365 Developer
Responsibilities:

- Designed and implemented scalable AI-centric solutions in hybrid cloud platforms (AWS SageMaker, Azure ML, Azure OpenAI Service & GCP Vertex AI) using modular and microservices architectures for efficient development and deployment.
- Highly skilled Python Application Developer with a strong background in Generative AI/ML and full-stack JavaScript development, eager to contribute to innovative AI-driven solutions.
- Possessing comprehensive expertise across Generative AI/ML, Python, and full-stack JavaScript, exceptionally well-positioned to develop and enhance cutting-edge applications. E.g.: const app = express(); app.listen(3000);
- Developed and implemented generative AI models in Python applications, leveraging NumPy for efficient data manipulation and core numerical operations, alongside TensorFlow, PyTorch, and Hugging Face for tasks such as text synthesis, image generation, and predictive modeling. E.g.: import numpy as np; data_tensor = np.random.rand(64, 128, 256) # Illustrates creating multi-dimensional arrays for model input

- Designing, building, and maintaining the scalable and secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, AWS Bedrock for hosting RAG pipelines, vector databases, and LLMs) to support advanced AI capabilities.
- **RAG Application Development**: Developed RAG-based applications utilizing platforms such as Azure OpenAI and Amazon Bedrock, alongside services like Cognitive Search. Possess a strong understanding of vector databases, including Amazon OpenSearch, Azure Search, and MongoDB Atlas.: results = vector_db.query(query_vector, top_k=5)
- Develop & maintain AI solutions for Aged Care APS use-cases, embedding AI applications into current business systems.
- Developed and deployed deep learning models leveraging frameworks such as PyTorch and TensorFlow. E.g.: model = tf.keras.Sequential([tf.keras.layers.Dense(10, input_shape=(784,))]) PyTorch E.g.: model = torch.nn.Linear(784, 10)
- Leveraged Hugging Face Transformers for Natural Language Processing (NLP) projects, fine-tuning pre-trained models like BERT and GPT for specific tasks including classification and summarization.
- Develop medical-tailored LLMs (BERT, GPT-4o, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline local AI systems (PrivateGPT-X, Ollama 4.0, Google NotebookLM, etc.) for data analysis, replacing inefficient tools like Jira or Confluence.
- Developed AI APIs on a secure, isolated NVIDIA Jetson Orin AGX 64GB Developer Kit, employing unsupervised learning for pattern discovery in unlabelled data & supervised KNN within Python for reinforcement learning tasks. Expertly selected and implemented algorithms (KNN, CNN, RNN, ANN, GAN) based on data characteristics and interpretability requirements.
- Developing Agent App streamlining Aged Care Sales, using Google's newly announced Agent-to-Agent (A2A) protocol for inter-agent communication & Model Context Protocol (MCP) for seamless integration with external tools (web search APIs).
- **Python Application Developer: AI & Generative Models:** Developed and deployed full-stack web applications leveraging Python frameworks (Django/Flask) for robust backend services, e.g., python manage.py runserver.
- Developed and deployed robust web applications using Python frameworks like Django and Flask, and crafted efficient web solutions with Next.js, leveraging server-side rendering and advanced application architecture. E.g.: python manage.py runserver (Django), app.run(debug=True) (Flask), and next dev (Next.js)
- Application Development & Backend APIs: Demonstrated ability to design and develop scalable and performant applications, including robust backend APIs and seamless system integrations.: data = session.query(User).filter_by(status='active').all()
- Built dynamic and responsive user interfaces using React.js and Redux, complemented by UI libraries like Tailwind and Ant Design. E.g.: const mapStateToProps = state => ({ count: state.counter.count }); followed by import { Button } from 'antd'; const MyComponent = () => <Button type="primary">Click Me</Button>;
- Experienced developing, maintaining, and optimizing cloud-based applications ensuring high availability, scalability, and cost-effectiveness. aws_lambda_client.update_function_configuration(FunctionName='MyService', Memory=512) Developed, optimized, & secured applications, integrating zero-trust principles & security best practices to ensure reliability, scalability, and compliance. E.g: response.headers['Strict-Transport-Security'] = 'max-age=31536000; includeSubDomains'
- Proficient in modern development workflows, utilizing platforms like n8n for workflow automation, and MCP RAG for retrieval-augmented generation, alongside Git for version control and robust CI/CD practices for continuous integration and deployment. E.g.: n8n start (n8n), git push origin main (Git/CI/CD), mcps-rag run --model gpt4 (MCP RAG), and docker build -t myapp:latest . && docker push myapp:latest (CI/CD)
- Developed and optimized data pipelines for AI model inference, enabling real-time predictions from large datasets, e.g., model.predict(new_data_batch).
- Proficient in Infrastructure as Code (IaC), leveraging tools like Bicep or Terraform for efficient and consistent resource provisioning. E.g.: resource storage 'Microsoft.Storage/storageAccounts@2023-01-01' = { name: 'mystorageaccount' }
- Collaborated with solution architects to design and develop fit-for-purpose AI solutions, adhering to ISM controls and zero-trust security architecture.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Python Application Developer/Software Engineer/Snr DevOps/Data Engineer/AI/ML/SP/D365 Developer
Responsibilities:

- Results-driven Python Application Developer with expertise in Generative AI-driven applications and a strong track record of engineering high-performance, data-driven solutions for complex financial trading environments.
- Comprehensive expertise across Generative AI/ML, Python, and full-stack JavaScript, uniquely positioned to design, develop, and optimize cutting-edge algorithmic trading systems and analytical applications. E.g., df['SMA_20'] = df['Close'].rolling(window=20).mean()
- Engineered and deployed high-performance Generative AI/ML models within Python applications, leveraging NumPy for optimized numerical operations and utilizing TensorFlow, PyTorch, and Hugging Face for advanced predictive modeling and synthetic data generation in financial contexts. E.g., sentiment_pipeline = pipeline("sentiment-analysis"); result = sentiment_pipeline("Financial markets are volatile.");
- Application Development & Backend APIs: Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 and Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations. **API Development**: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
  System Integration: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.

- Engineered scalable, high-performance Python applications and robust backend APIs, facilitating seamless cloud-native integrations across Azure/AWS services for real-time data processing and AI solution enablement. E.g., sqs_client.send_message(QueueUrl=queue_url, MessageBody=json_data)
- Architected and implemented Retrieval-Augmented Generation (RAG) solutions leveraging LLMs on platforms like Azure OpenAI and Amazon Bedrock, integrating with enterprise search services and vector databases (e.g., Amazon OpenSearch, Azure Search, MongoDB Atlas) to provide real-time, context-aware financial intelligence and risk insights.
  E.g.: relevant_financial_docs = vector_store.similarity_search(query_embedding, k=5)
- Developed and maintained advanced AI solutions, including LLM-powered applications, specifically tailored to RBA APS (Australian Prudential Regulation Authority standards) compliance and embedded these into core financial trading and business systems to enhance operational efficiency and regulatory adherence.
  E.g.: compliance_check_result = internal_ai_service.assess_trade_for_aps(trade_data)
- Developed and deployed high-performance deep learning models using PyTorch and TensorFlow for critical financial applications such as market prediction, anomaly detection in trading data, and algorithmic strategy optimization.
  E.g.: (TensorFlow): model.fit(training_data, epochs=10, validation_split=0.2)
  E.g.: (PyTorch): output = model(input_tensor); loss = criterion(output, target)
- Leveraged Hugging Face Transformers for Natural Language Processing (NLP), fine-tuning pre-trained LLMs (e.g., BERT, GPT) for critical financial tasks including sentiment analysis of market news, regulatory document classification, and earnings call summarization. E.g.: sentiment_score = classifier(financial_news_text)[0]['score']
- **MLOps & Pipeline Automation**: Proficient in MLOps tools, including AWS SageMaker, for implementing robust monitoring frameworks and automating machine learning pipelines. E.g.: sm_estimator.fit({'training': s3_data_path})
- Integrated Python-based AI solutions with JavaScript frontends to deliver comprehensive and interactive web applications.
- Developed efficient web solutions leveraging Next.js for server-side rendering and advanced application architecture, e.g., getServerSideProps for data fetching, or const router = useRouter(); for client-side navigation.
- Utilized Node.js to build scalable, multi-threaded, or clustered applications, enhancing backend capabilities, e.g., const cluster = require('cluster'); for process management, or new Worker('./worker.js') for background tasks.
- Ensured seamless user experiences through strong integration capabilities between front-end and back-end systems, e.g., axios.post('/api/data', payload) for API calls, or useEffect(() => { /* fetch data */ }, []) for data loading on component mount.
- Developed efficient web solutions leveraging Next.js for server-side rendering and advanced application architecture, e.g., getServerSideProps for data fetching.
- Utilized Node.js to build scalable, multi-threaded, or clustered applications, enhancing backend capabilities, e.g., const cluster = require('cluster');.
- Ensured seamless user experiences through strong integration capabilities between front-end and back-end systems, e.g., axios.post('/api/data', payload).
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Developed Mulesoft Anypoint APIs and Dashboards to provide real-time insights and monitoring of trading activities.
- Full-stack engineer skilled in Angular/TypeScript for frontend development, Python/Flask or Django for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and web frameworks (e.g., Flask/Django) to ensure seamless data flow using RESTful web services.
- Develop Python web application components using frameworks like Flask or Django.
- Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with RBA's enterprise services layer, databases, and host systems.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Python Application Developer /Snr DevOps/Data Engineer/Java/Mulesoft/SharePoint/D365 Developer
Responsibilities:

- Results-oriented Python Application Developer with a strong quantitative background, skilled in creating high-performance, data-driven solutions. Proven expertise in Python and full-stack JavaScript, uniquely positioning me to design, develop, and optimize cutting-edge machine learning applications.
- Leveraged these skills to accurately predict the likelihood of international student enrollment based on various factors using trained machine learning models. E.g.: predicted_students = predict_enrollment(prospective_students.copy())
- Engineered and deployed high-performance Machine Learning models within Python applications, leveraging Azure Machine Learning Studio's capabilities with R and Python support for advanced predictive modeling and synthetic data generation, significantly impacting financial applications. E.g., model_output <- predict(trained_model, newdata)
- **Front-End Development (Next.js)** : Engineered dynamic web interfaces for student enrollment using Next.js, ensuring a seamless user experience and optimized data loading for complex forms and dashboards. E.g.,
  const { data } = await useSWR('/api/courses'); (Client-side data fetching for dynamic content)

- **Back-End Development (Node.js)**: Developed robust and scalable backend services with Node.js to manage high volumes of concurrent enrollment applications, secure data transactions, and integrate with university systems. E.g., app.post('/apply', (req, res) => { /* process async data */ }); (Handling asynchronous enrollment submissions)
- **Front-End Development**: Developed intuitive and responsive user interfaces for the UC student application portal using React.js and Redux, enhanced by modern UI libraries for a seamless user experience. E.g., dispatch({ type: 'UPDATE_APPLICANT_STATUS', payload: 'submitted' }); (Manages application state updates)
- **Cloud Application Management**: Managed and optimized cloud-based applications for the student enrollment system, ensuring high availability, scalability, and cost-effectiveness on leading cloud platforms. E.g., aws s3 sync local_data/ s3://uc-student-data-bucket/ --delete (Ensures data consistency and efficient storage)
- **Application Security & Optimization**: Secured and optimized critical application functionalities by integrating zero-trust principles and best practices to ensure reliability, scalability, and compliance for sensitive student data. E.g., const helmet = require('helmet'); app.use(helmet()); (Implements essential security headers)
- Application Development & Backend APIs with Mulesoft Anypoint Platform
- Demonstrated ability to design and develop scalable and performant applications, including robust backend APIs and seamless system integrations.
    - For API Development: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
    - For System Integration: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.
- Built Dynamic and Responsive User Interfaces: Built dynamic and responsive user interfaces using React.js and Redux, complemented by UI libraries like Tailwind and Ant Design.
    - For React/Redux State Management: dispatch({ type: 'ADD_ITEM', payload: newItem })
    - For UI Libraries (Tailwind/Ant Design):
- Application Development & Backend APIs: Demonstrated ability to design and develop scalable and performant applications, including robust backend APIs and seamless system integrations.: data = session.query(User).filter_by(status='active').all()
- Built dynamic and responsive user interfaces using React.js and Redux, complemented by UI libraries like Tailwind and Ant Design. E.g.: const mapStateToProps = state => ({ count: state.counter.count }); followed by import { Button } from 'antd'; const MyComponent = () => <Button type="primary">Click Me</Button>;
- Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- Designed and maintained robust CI/CD pipelines using platform-specific tools:
    1. **Azure: Azure DevOps**
    2. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    3. **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Building microservices applications using Python & JavaScript with frameworks like Flask & Node.js. Expertise in consuming RESTful web services to integrate with UC's enterprise services layer, databases, and host systems.
- **Automated Kubernetes Deployments:** Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or even on-premises environment.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Python Application Developer /DevOps Engineer/Data Engineer /Software Test Engineer
Responsibilities:

- Data-driven Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. My proficiency in Python and full-stack JavaScript allows me to design, develop, and optimize advanced machine learning applications that deliver actionable insights.
- Applied these capabilities to develop machine learning models that accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain. E.g.: high_risk_claims = claims[model.predict_proba(features)[:, 1] > 0.7]
- Machine Learning Models: Developed and deployed advanced predictive models using Python and Azure Machine Learning, optimizing claims processing efficiency and enhancing fraud detection capabilities. E.g, model_output = trained_model.predict(new_claim_features)

- Front-End Development: Designed and implemented intuitive web interfaces with Next.js, streamlining the user experience for complex forms and dashboards crucial to efficient injury compensation claim management.
- E.g, const { data: claimData } = await useSWR('/api/claim-details');
- Back-End Development (Node.js): Engineered secure and scalable Node.js backend services to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems. E.g., app.post('/api/claims', async (req, res) => { /* secure claim intake */ });
- Front-End Development (React.js & Redux): Developed intuitive and responsive user interfaces using React.js and Redux, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows. E.g., dispatch(setClaimStatus('Submitted'));
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed AWS Redshift with its built-in data processing capabilities for near-real-time processing & analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using API Gateway
- Deploy and maintain web applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.

- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~10, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Automation Engineer – Westpac – FinXL – 14/7/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Automation Engineer/Integration Solution Architect/Snr DevOps/Data Engineer/Snr Network Engineer /Java/.Net Python /AI/ML/BI/MuleSoft/BizTalk/SharePoint/D365 Developer/EDW –Japan (7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**

Gov. Health: Senior Automation Engineer/Snr Network/DevOps/Data Engineer/Python/AI/ML/SP/D365 Developer/EDW
Responsibilities:

- Architected and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Configured encrypted tunnels for secure data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with AWS CloudWatch.
- Integrated M365 environments (O365, MS Teams, SharePoint online, Dynamics 365, Power Automate,Power App, Power BI) & AI Builder for D365 into the AWS Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- Professionally leveraging AWS services (S3 for scalable storage, EC2 for compute, and managed container services like EKS and ECS Fargate for scalable application deployment). Designed and implemented robust big data processing workflows and scalable ETL/ELT data pipelines using Apache Airflow and Cloud Native Distribution Infrastructure (CNDI). These pipelines incorporate advanced analytics tools such as Spark, Presto, and Hive to derive actionable insights.
- Designed and implemented end-to-end automation solutions using Python and Ansible (Tower, AWX, Red Hat AAP 2.5), developing custom modules and plugins to streamline infrastructure provisioning and configuration across hybrid cloud, on-prem, and Red Hat Enterprise Linux (RHEL9/10) environments. E.g., ansible-playbook provision_rhel_app_servers.yml
- Automated infrastructure and application deployments using Ansible and Terraform in Kubernetes/OpenShift clusters, ensuring scalable and repeatable workflows across cloud-agnostic platforms. E.g., terraform apply -auto-approve && ansible-playbook deploy_app_on_openshift.yml
- Built automated Jenkins pipelines leveraging Jenkinsfile Runner via GitHub Actions to support comprehensive automated testing, robust deployment, and infrastructure updates in multi-environment hybrid cloud setups. E.g., gh workflow run 'ci-cd-pipeline.yml' --ref main
- Developed robust Python scripts and API integrations to automate interactions between infrastructure, databases (e.g., MongoDB), and third-party platforms, improving operational efficiency and reducing manual intervention.
- Automated the creation of secure, scalable Infrastructure-as-Code (IaC) CI/CD pipelines, built on secure, isolated Generative AI platforms utilizing LLMs (OpenAI, Copilot, Gemini, Perplexity, & self-hosted models deployed on OLLAMA4.0 within secure private virtual networks), MCP+RAG architectures, and n8n agentic workflows—to enable robust deployment of Python web applications (Django, Flask) and high-performance Next.js solutions with server-side rendering across hybrid cloud environments (Azure, AWS, GCP). E.g., curl -X POST https://n8n.yourdomain.com/webhook/generate-cicd-pipeline
- Designed and implemented agent-to-agent and n8n-based orchestration agentic workflows (integrating tools like LangChain) to significantly streamline CI/CD processes (GitHub Actions, GitLab CI/CD), enhancing code review efficiency, robust automated testing, and secure deployments for complex web architectures leveraging both Python and Next.js frameworks in multi-cloud settings. E.g., curl -X POST https://n8n.yourdomain.com/webhook/trigger-smart-cicd-agent
- Leveraged RAG (Retrieval-Augmented Generation) and AI-powered agents to automate Pull Reqest analysis, pipeline debugging, and dynamic infrastructure provisioning, accelerating delivery cycles and ensuring compliance for enterprise-grade web applications in aged care financial management. E.g.: n8n start (n8n), git push origin main (Git/CI/CD), mcps-rag run --model gpt4 (MCP RAG), and docker build -t myapp:latest . && docker push myapp:latest (CI/CD)
- Designed agent-to-agent and n8n-based orchestration workflows to streamline multi-cloud DevOps processes, integrating advanced Generative AI tools to automate the build and deployment of Next.js solutions with server-side rendering and modular architecture, ensuring resilience, compliance, and performance across aged care financial management systems.
- Proficient in modern development workflows, utilizing platforms like n8n for workflow automation, and MCP RAG for retrieval-augmented generation, alongside Git for version control and robust CI/CD practices for continuous integration and deployment. E.g.: n8n start (n8n), git push origin main (Git/CI/CD), mcps-rag run --model gpt4 (MCP RAG), and docker build -t myapp:latest . && docker push myapp:latest (CI/CD)
- Developed and optimized data pipelines for AI model inference, enabling real-time predictions from large datasets, e.g., model.predict(new_data_batch).
- Proficient in Infrastructure as Code (IaC), leveraging tools like Bicep or Terraform for efficient and consistent resource provisioning. E.g.: resource storage 'Microsoft.Storage/storageAccounts@2023-01-01' = { name: 'mystorageaccount' }
- Collaborated with solution architects to design and develop fit-for-purpose AI solutions, adhering to ISM controls and zero-trust security architecture.
- Proactively troubleshot and resolved complex automation issues across Linux-based systems, ensuring high availability and performance of automation pipelines in cloud-agnostic environments. Delivered reusable, modular automation frameworks that integrated with diverse systems and tools via REST APIs, enabling seamless orchestration across hybrid infrastructure layers. E.g., dmesg -T | grep -Ei 'error|fail|warn|critical|oom'
- **Network Engineering**: Led design and deployment of robust network solutions, leveraging deep knowledge of CIDR, TCP/IP stack, OSPF, RIP, and BGP for optimal performance & security across diverse WAN & hybrid cloud environments.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Automation Engineer/Snr Network/DevOps/Data Engineer/Java/Python/AI/ML/SP/D365/BI Developer
Responsibilities:

- Specialized in designing and implementing end-to-end automation solutions using Python and Red Hat Ansible Automation Platform (including Ansible Tower, and Ansible Web eXecutor(AWX), developing custom modules and plugins to streamline secure, low-latency infrastructure provisioning and configuration for RBA financial trading platforms across hybrid cloud, on-prem, and RHEL9 environments. E.g., ansible-playbook apply_rba_security_baseline.yml
- Automated resilient infrastructure and real-time application deployments using Ansible and Terraform within Kubernetes/OpenShift clusters, ensuring scalable and repeatable workflows for mission-critical financial trading services across cloud-agnostic platforms. E.g., terraform apply -auto-approve && ansible-playbook deploy_low_latency_trading_engine.yml
- Accelerated RBA Trading Operations: Designed and implemented automated Jenkins CI/CD pipelines, integrated with GitHub Actions, to facilitate rapid, reliable, and secure deployment of trading applications and infrastructure updates within multi-environment hybrid cloud setups. E.g., kubectl apply -f deployment.yaml
- Optimized Financial Data Workflows: Created robust Python scripts and API integrations, automating interactions between critical trading infrastructure, databases (e.g., MongoDB), and external financial platforms, thereby enhancing operational efficiency and data integrity.
- Streamlined RBA Financial Infrastructure: Drove efficiency through Infrastructure as Code (IaC) practices, utilizing Bicep and Terraform to define, deploy, and manage consistent infrastructure across RBA's diverse financial trading ecosystems. E.g., terraform apply -auto-approve
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Developed and implemented robust CI/CD pipelines using GitHub Actions to streamline Mulesoft application deployment. Leveraged YAML scripting and automation to enhance RBA development efficiency and reduce risk.
- **CloudWatch Observability Expertise:** Implemented comprehensive CloudWatch observability solutions, including metrics, logs, and dashboards. Leveraged CloudWatch Insights for advanced log analysis and troubleshooting.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- **Cloud Infrastructure Engineering**: **Azure Expertise**: Proven proficiency in designing, implementing, and managing Azure infrastructure, including VNets, subnets, NSGs, SSO, FortiGate Firewalls, FortiClient VPN with MFA, and Azure AD-based access control for Azure Key Vault.
- **Network Engineering**: Engineered and optimized intricate network topologies, demonstrating expertise in CIDR, TCP/IP, OSPF, RIP, BGP, ISIS, BFD, LDP, and L2TP protocols to deliver reliable, scalable WAN & hybrid cloud solutions.
- Deployed and managed IP and MPLS networks connectivities between RBA and all major financial institutions
- **SD-WAN**: Experienced in leveraging SD-WAN solutions to optimize WAN performance and security.
- **Cloud Operations**: Managed & configured Azure cloud services, ensuring high availability, resiliency, & cost-efficiency. Supported development & test environments and associated DevOps tools (Git, Ansible, Jenkins, Docker, etc.).
- Utilized Terraform to provision the infrastructure (VMs, networks, ARM, etc.), and then used PowerShell Desired State Configuration (DSC) to configure the operating systems and applications on those machines. This hybrid approach allows you to manage both infrastructure and configuration in a consistent and automated way, aligning well with IaC principles.
- Implemented robust CI/CD pipelines with Mulesoft to automate application delivery and reduce manual intervention. Leveraged Terraform to efficiently provision and manage cloud infrastructure resources (Azure, AWS, GCP) for Mulesoft Anypoint API deployments connecting to major financial institutions 24x7.
- **Networking Proficiency**: Extensive experience with Cisco networking technologies.
- Designed, developed, and maintained automated data pipelines and API orchestration using either:
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Optimized data pipelines utilizing Airflow and Luigi, reducing processing time by 30% and improving resource utilization.
- Automated model versioning and rollbacks with MLflow, enabling seamless rollbacks and mitigating production issues: e.g. mlflow.sklearn.log_model(model, "linear_model") & rollback by client.transition_model_version_stage()
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- **Integration Solution Design with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)

Gov. University: Snr Automation Engineer/Snr Network/DevOps/Data Engineer/Python/ML/SharePoint/D365 Developer
Responsibilities:

- Cloud Integration Solution Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow
- Enhanced Automation for University IT: Proficient in designing and implementing comprehensive automation solutions with Python and Ansible Tower, including the development of custom modules and plugins, to streamline secure infrastructure provisioning and configuration for the University of Canberra's IT systems, particularly those supporting international student services, across hybrid cloud and RHEL environments. E.g., python migrate_student_data.py
- Resilient System Deployment for Student Services: Orchestrated automated, resilient infrastructure and application deployments leveraging Ansible and Terraform within Kubernetes/OpenShift clusters, guaranteeing scalable and repeatable workflows for vital University services, such as the international student enrolment system, across diverse cloud platforms. E.g., oc apply -f student-api-deployment.yaml
- Resilient System Deployment for Student Services: Orchestrated automated, resilient infrastructure and application deployments leveraging Ansible and Terraform within Kubernetes/OpenShift clusters, guaranteeing scalable and repeatable workflows for vital University services, such as the international student enrolment system, across diverse cloud platforms. E.g., oc apply -f student-api-deployment.yamlProvided technical supporting and troubleshooting cloud resources to ensure availability, resiliency and costing efficiency.
- Strategic planning regarding optimizations for the improvement of the core infrastructure.
- **Networking Proficiency**: Extensive experience with Cisco networking technologies
- **Cloud Infrastructure Engineering**: Azure expert, proficient in designing, implementing, and managing Azure infrastructure, including VNets, subnets, NSGs, firewall rules, SSO, FortiGate, FortiClient VPN with MFA, and Azure AD-based access control for Azure Key Vault.
- Demonstrated expertise in integrating Terraform and Mulesoft to automate complex API deployments and infrastructure provisioning.Successfully designed and implemented CI/CD pipelines using Terraform and Mulesoft, optimizing application lifecycle management.Applied Terraform to provision and manage cloud infrastructure resources (Azure, AWS) with a focus on efficiency, security, and scalability.
- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
- Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Cloud Infrastructure Engineer/Senior Network/DevOps Engineer/Clouds/Linux Administrator
Responsibilities:

- Skilled Cloud Integration Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow
- **Network Engineering**: Strong understanding of the interplay between CIDR, the TCP/IP stack, and routing protocols (IGPs: OSPF, RIP; EGP: BGP) in WAN and hybrid cloud environments.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python with ARM templates, Terraform, or Ansible. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration in Azure, & equivalent container orchestration services in AWS and GCP (e.g., ECS Fargate, GKE).
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Proficient in Linux system administration, including kernel optimization (sysctl, vmstat), performance tuning (iostat, sar), and security hardening (iptables, fail2ban). Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all EML's data pipelines.

- Managed the integration of system solutions across various platforms and technologies, supporting DevOps and Cloud.
- Optimized the pipeline for performance, cost, and efficiency by applying best practices such as resource allocation, scaling, health checks, and logging in addition to partitioning, indexing, and optimizing the ADF pipeline structure.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Streamlined DevOps processes by implementing robust version control practices with Git, including branching strategies, code reviews, and merge requests. Utilized configuration management tools to consolidate settings and streamline infrastructure management, ensuring consistent provisioning.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Cloud Infrastructure Engineer/Snr Network/DevOps Engineer/SharPoint Developer
Responsibilities:

- Proven track record of designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- Streamlined cloud operations by automating resource provisioning and management using infrastructure as code (IaC). Proficient in configuring and managing hybrid cloud environments (Azure, AWS, GCP) across development, testing, and production, leveraging tools like Ansible, CloudFormation, boto3, and Pulumi.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: System Administrator /Senior Systems Engineer /Administrator (AD/SCCM/SCOM)/DBA
Responsibilities:

- Managed and maintained Azure AD, D365 CRM, SharePoint, and M365 for a large organization, supporting over 1000 users.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Implemented JIRA project management tool, streamlining issue tracking and collaboration across cross-functional teams.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Network Administrator /Systems Engineer /Wintel Systems Administrator (AD/SCCM/SCOM) /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.

- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Designed and maintained a robust and secure network infrastructure, supporting data and voice communication for 1200+ employees.
- Implemented advanced firewall configurations and intrusion prevention systems, safeguarding against cyber threats.
- Conducted regular network performance monitoring and optimization, enhancing network efficiency by 20%.
- Designed and managed network infrasctructure and security measures.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Senior Network Engineer/Senior Systems Engineer/Integration Developer /Data Engineer**
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, and infrastructure products on HP G3-G6 servers, HP and Dell Blade Servers, Dell and Citrix Desktops, & HP/NetApp/Dell storage solutions.
- Led data center infrastructure planning and management, ensuring the availability and reliability of critical business applications. Provided network design for BAU operations and technical support to the US Head Office SOC Team.
- Deployed and managed IP and MPLS network connectivity between MS global data centers and branches.
- Worked in the NOC team to perform large-scale Windows Server auto-installation using PXE scripts on Active Directory-linked Altiris imaging servers and auto-install Windows desktop (XP/Vista) OS using MDT 2008/BDD 2007 Update 2 deployment best practices
- Implemented SCCM for automated software deployments and patch management, reducing manual efforts by 40%.
- Provided technical guidance to junior team members and collaborated with cross-functional teams on IT projects.
- Designing and architecting SCCM2007R2 for enterprise deployment, patch and update management solution
- Plan, design and manage Active Directory with high level support to AD Migrations - within and between AD forests
- **Networking Proficiency**: Extensive experience with Cisco networking technologies
- **Network Engineering**: Understood and configured complex network environments, leveraging knowledge of CIDR, the TCP/IP stack, and routing protocols (IGPs: OSPF, RIP; EGP: BGP) in both WAN and hybrid cloud scenarios.
- Design, implement, configure and administer Infrastructure solutions for IaaS, Saas, PaaS etc.
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Assisted in managing IT infrastructure and ensuring optimal performance.
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers& F5 load balancers v9,
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Administration of Windows Server2003~2008R2, SQL Server2005~2008, AD, DNS, DHCP, Exchange2007, MOSS2007
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities. Promoting & improving incident & problem management, change / release management policies & procedures by SCOM

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, AIX/Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~10, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.

- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
- **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

DevOps Engineer (On-Prem) –AKS – Parramatta - Companion Systems – 6/July/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Snr DevOps Engineer/Integration Solution Architect /Software Engineer /Snr Data Engineer/AIX/Unix/Linux Administrator/ Java/.NET/Python/AI Automation/DL/ML/BI/SharePoint/D365 Developer /EDW /DBA-Japan(7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr DevOps Engineer/Integration Solution Architect /Snr Data Engineer/Py/AI/ML/SharePoint/D365 Developer
Responsibilities:

- Architected and automated on-premise application deployments via Azure DevOps Server, leveraging Git and agile development methodologies alongside AI-assisted C#.NET Core code generation (e.g., Workik AI), containerization (Docker, Kubernetes), and AI-enhanced Infrastructure-as-Code (Terraform, Ansible) across diverse Linux and virtualized server landscapes, which significantly enhanced deployment efficiency and system reliability.
- Leveraged Azure DevOps Server as the central orchestration platform for .NET development lifecycles, incorporating AI-generated Docker image creation pipelines and deployment to on-premise registries and Kubernetes clusters provisioned
- Managed and optimized Azure DevOps Server build pipelines on on-premise agents to compile AI-assisted .NET code, execute unit tests, and generate web deploy packages, streamlining CI processes with intelligent code generation and robust automation. Work with various teams to improve the scalability, reliability, and performance of their systems.
- Orchestrated automated deployments to on-premise IIS servers using Azure DevOps Server release pipelines, leveraging built-in tasks and custom, AI-generated PowerShell scripts for efficient file transfer and IIS application pool configuration.
- Proven Azure DevOps Engineer with a strong "Automation First" mindset and capability to review and uplift existing Azure infrastructure and deployment workflows, with a primary focus on identifying and implementing comprehensive automation solutions leveraging Terraform, Python/Bash, Azure DevOps, Docker, and AKS. E.g.: Terraform: resource "azurerm_kubernetes_cluster" "aks" { name = "my-aks" location = "australiaeast" resource_group_name = "rg-aks" }
- **Azure DevOps**: pipeline = azurerm.Pipeline(name="app-deploy", project=project.id, yaml_path="app-deploy.yaml")
- **Docker**: docker build -t my-app:latest . **Azure Kubernetes Service (AKS)**: kubectl apply -f deployment.yaml
- Proficient in Azure DevOps, Azure, Git, Terraform, YAML, and various scripting/programming languages (Python, PowerShell, C#, JavaScript).
- **Azure Functions**: Developed serverless APIs and event-driven processes using Azure Functions to automate real-time data ingestion and workflow orchestration, reducing processing time by 30%. E.g: public static async Task<IActionResult> Run([HttpTrigger(AuthorizationLevel.Function, "post")] HttpRequest req)
- **Azure App Services**: Engineered scalable web applications and RESTful APIs deployed on Azure App Services, enabling seamless integration with front-end platforms and external systems. E.g: services.AddControllers().AddNewtonsoftJson(options => { options.SerializerSettings.ReferenceLoopHandling = Newtonsoft.Json.ReferenceLoopHandling.Ignore; });
- **Azure OpenAI for AI-Driven Risk Assessment**: Implemented Azure OpenAI solutions for AI-driven risk assessment in regulatory compliance, extracting critical insights and identifying potential anomalies from large volumes of unstructured text data. E.g: response = openai_client.chat.:completions.create(model="gpt-4", messages=[{"role": "user", "content": "Analyze risk in audit report:..."}])
- **Azure Cognitive Search with Cosmos DB**: Designed and deployed high-performance search solutions using Azure Cognitive Search for rapid information retrieval from regulatory documents, leveraging Cosmos DB as a highly available and globally distributed data source..E. g: SearchClient searchClient = new SearchClient(serviceEndpoint, indexName, credential); SearchResults<MyDocument> results = searchClient.Search<MyDocument>("compliance breach");
- Linux Server Provisioning: Leveraged Ansible playbooks (e.g., vmware_guest.yml, network_config.yml, package_install.yml, deploy_app.yml) for automated VMware/Hyper-V VM creation, network configuration, package installation, and application deployment, often integrated into CI/CD pipelines via Azure Pipelines (in Azure DevOps Server) or Bamboo CI.
- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline local AI Agents (Ollama 4, Google NotebookLM) till MCP using Agent2Agent portocol) for data analysis and decision making.
- Designed and implemented AI-driven workflows using n8n to automate client intake, medical history triage, and wellness reporting for aged care service, integrating AI agents & LLMs to enhance operational efficiency & personalized aged care.

- Developed AI APIs on a secure, isolated NVIDIA Jetson Orin AGX 64GB Developer Kit, employing unsupervised learning for pattern discovery in unlabelled data & supervised KNN within Python for reinforcement learning tasks. Expertly selected and implemented algorithms (KNN, CNN, RNN, ANN, GAN) based on data characteristics and interpretability requirements.
- Developing Agent App streamlining Aged Care Sales, using Google's newly announced Agent-to-Agent (A2A) protocol for inter-agent communication & Model Context Protocol (MCP) for seamless integration with external tools (web search APIs).
- Integrated Power Platform (Power Automate, Power Apps, Power BI) using Custom Connectors for secure API interaction with on-premise SharePoint (2019/2022), Dynamics 365, SQL Server 2019/2022 and cloud services.
- Leveraged Azure DevOps Server for source control and CI/CD pipelines to manage deployments to cloud services like Azure API Management, Azure Container Apps, and Azure Spring Apps.
- Developed Power Apps Component Framework (PCF) controls for enhanced visualization of external CRM data within model-driven applications. Built Power Platform solutions enhanced with AI Builder capabilities.
- Technical support for aged care business applications covering SharePoint 2019/2022/Dynamics 365, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr DevOps Engineer/Integration Solution Architect/Snr Unix/Linux Engineer/Py/ML/D365/SP Developer
Responsibilities:

- Architected and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Configured encrypted tunnels for secure data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with AWS CloudWatch.
- Integrated M365 environments (O365, MS Teams, SharePoint online, Dynamics 365, Power Automate,Power App, Power BI) & AI Builder for D365 into the AWS Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- Implemented robust CI/CD pipelines to automate secure application delivery across diverse infrastructure (on-premise Azure DevOps Server/TFS; cloud-based Azure, AWS, GCP), utilizing Git and agile principles to accelerate deployment cycles for performance-critical systems.
- Designed and implemented modern cloud solutions on the Azure stack in close collaboration with RBA financial services stakeholders, employing Azure Functions and App Services to automate essential financial workflows. The integration of OpenAI for intelligent document processing and Cognitive Search for quick retrieval of financial compliance data, supported by the scalability of Cosmos DB, dramatically enhanced operational efficiency and intelligent automation.
- **Designed and implemented Medallion Architecture with Data Lake in Azure:**
- **Batch Layer:** Utilized Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. The Bronze layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implemented Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. The Silver layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leveraged Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. The layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- **Azure Functions**: Implemented secure, serverless event processing for RBA financial data using Azure Functions. E.g.: [FunctionName("ProcessFinancialData")] public static async Task Run([QueueTrigger("rba-financial-queue", Connection = "StorageAccount")] string financialRecord)
- **Azure App Services**: Developed secure & scalable RESTful APIs on Azure App Services for RBA financial data exchange: builder.Services.AddAuthentication(JwtBearerDefaults.AuthenticationScheme).AddMicrosoftIdentityWebApi(builder.Configuration.GetSection("AzureAd"));
- **Azure OpenAI for AI-Driven Compliance Analysis**: Leveraged Azure OpenAI to extract key insights and identify potential compliance risks from unstructured RBA financial documents. E.g.: extracted_insights = openai.Completion.create (engine="text-davinci-003", prompt="Summarize key compliance risks in the following RBA report: ...")
- Azure Cognitive Search for Financial Risk Intelligence: Built high-performance search functionality with Azure Cognitive Search to rapidly locate risk-related information within RBA financial documents indexed from Cosmos DB. E.g.: SearchResults riskFindings = searchClient.Search("market risk", new SearchOptions { Select = new[] { "title", "summary", "documentId" } });
- Proficient in applying Zero Trust principles via Azure IAM and Key Vault to secure identities and privileged access, while utilizing Microsoft Sentinel and SIEM/SOAR technologies for AI-powered anomaly detection, fraud prevention, and rapid incident response. Demonstrated alignment with PCI DSS, ISO 27001, and global financial security regulations across network, workstation, and cloud environ
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime

- Developed, constructed, and implemented bespoke Power Apps (Canvas and Model-driven) applications, alongside automating workflows via Power Automate and establishing secure data structures and entities in Dataverse, further enhanced by the creation and upkeep of insightful Power BI dashboards and reports.
- Proficient in applying Zero Trust principles via Azure IAM and Key Vault to secure identities and privileged access, while utilizing Microsoft Sentinel and SIEM/SOAR technologies for AI-powered anomaly detection, fraud prevention, and rapid incident response. Demonstrated alignment with PCI DSS, ISO 27001, and global financial security regulations across network, workstation, and cloud environ
- Architected and implemented diverse APIs to facilitate connectivity between internal & external data platforms, using Azure services including Azure Functions, Logic Apps, Service Bus, & API Management for their construction and maintenance.
- Development of tailored Mulesoft solutions that align with business requirements for scalability, security, and efficiency. Guide teams in adhering to standards and governance frameworks, leveraging technical expertise in server-side languages (Java, Node.js, etc.) and front-end frameworks (React JS/Next JS, Mulesoft Anypoint API, etc.).
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer/Integration Solution Architect /Snr Data Engineer/Py/ML/SP/365 Developer/EDW
Responsibilities:

- Designed and implemented automated CI/CD pipelines for application deployment across both on-premise (leveraging Azure DevOps Server/TFS) and cloud (Azure DevOps, AWS CodePipeline/CodeBuild/CodeDeploy, GCP Cloud Build/Cloud Deploy) environments, integrating Git for version control and agile development principles.
- Managed and continuously optimized Azure DevOps Server (formerly TFS) build pipelines on dedicated on-premise agents, facilitating the compilation of .NET code, comprehensive unit testing, and MSBuild and Web Deploy via MSDeploy.exe for web deployment, significantly streamlining Continuous Integration (CI) processes and contributing to improved system scalability, reliability, and performance in collaboration with various engineering teams.
- Automated deployments to on-premise IIS servers leveraging Azure DevOps Server (formerly TFS) release pipelines, employing both native tasks and bespoke PowerShell scripts to ensure efficient file transfer and precise IIS application pool configuration. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration within Azure.
- Leveraged virtualized server landscapes, alongside Ansible for provisioning, resulting in a marked improvement in deployment efficiency and system reliability. Leveraged Infrastructure as Code (IaC) principles using Python with ARM templates to automate the provisioning and configuration of Azure network resources.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Teams/SharePoint 2016/Dynamics CRM 2016/Microsoft Flow).Implemented secure landing zones for cloud workloads using Azure Blueprints.
- Applied analytical techniques to proactively detect anomalies and mitigate fraud, enhancing security posture with expertise in Azure IAM and Key Vault.
- Python programming to design and implement automated and intelligently optimized pipelines for AWS IaC pipeline creation, prediction, and optimization using CloudFormation and boto3: cfn = boto3.client('cloudformation')...
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design and deploy APIs that seamlessly integrate AWS Lambda functions with Azure Logic Apps, enabling real-time data synchronization between the two platforms.Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- Designed and implemented custom solutions using Power Apps and Microsoft Flow, leveraging Mulesoft Anypoint API for scalable integrations and automation. Developed and deployed Azure Functions and Logic Apps to establish connectivity with external APIs, managed and secured through Azure API Management (APIM).
- Developed and maintained Machine Learning (ML) and Deep Learning (DL) models to enhance customer engagement and sales by efficiently manipulating and extracting insights from big data. Contributed to the adoption of MLOps practices.Drive collaboration with our Microsoft Partner to coordinate backlog of product enhancements by adopting MLOps.
- Developed custom components using Mulesoft Anypoint API to create interactive UI elements.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- Leverage Python programming with CyberArk SDK to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using Python with Pandas and Airflow to offer flexibility for building intricate CI/CD pipelines tailored to specific needs.
- Leveraged SharePoint for process integration and Drupal/HTML with JavaScript for form development, along with backend SQL Server 2014 database for efficient data retrieval and customer service operations in Dynamics CRM 2016.

- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- Drive collaboration with our Microsoft Partner to coordinate backlog of product enhancements by adopting MLOps.
- **Leverage Python programming with CyberArk SDK** to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
  # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer /Software Test Engineer/.NET/Python/SharePoint/D365 Developer/EDW
Responsibilities:

- Developed and implemented data integration solutions, including ETL/ELT processes, within the Microsoft ecosystem.
- Designed data models, loading approaches, and source-to-target mapping documentation to facilitate efficient data flow.
- Analyzed diverse data sources, types, and formats to define requirements for data transformation, loading, and modeling.
- Utilized Python for custom ETL/API orchestration, demonstrating adaptability across different application environments.
- Contributed to the implementation of automation frameworks and CI/CD pipelines.
- Applied DevOps principles to automate build and deployment processes for online applications.
- Leveraged Python programming skills for data-related tasks and automation.
- Worked with various database systems, including SQL, Oracle, PostgreSQL, MySQL, DynamoDB, and MongoDB.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Designed, developed, and maintained automated ETL/ELT data pipelines and API orchestration using either:
  - **Azure Synapse Analytics** with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - **AWS Glue** for data processing and integration with other AWS services to load data into target Aurora databases.
  - **GCP Cloud Dataflow** or **Cloud Data Fusion** for data processing and integration, and Google BigQuery for data warehousing and analytics.
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: DevOps Engineer /Data Engineer /Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.
- Executed ELT/ETL processes and advanced analytics using R and Python for data enrichment and analysis.
- Designed DWH concepts from Bill Inmon to Ralph Kimball and build data models using Toad Data Modeler 6.x
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
  - Azure: Azure DevOps, Azure Pipelines
  - AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
  - GCP: Cloud Build/Cloud Deploy
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior AIX/Unix/Linux Engineer/Data Engineer /BI Developer /DWH /DBA
Responsibilities:

- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Data Vault modelling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /AIX/Unix /Linux Engineer /BI Consultant /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 using **LAMP** Stack to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate process

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, OpenAI o1, GPT-3.5/4o, AI Agent, n8n ai agent, Azure OpenAI ML Services, Copilot, Google GEMINI Pro 1.5~2.5 NotebookLM; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase, **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

DevOps Engineer – Revenue NSW – 1/7/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Integration Solution Architect/Snr Data Engineer/Snr Clouds/AIX/Unix/Linux Administrator /EDW/ Snr Systems Engineer/Java/.NET/Python/AI/ML/BI/SharePoint /D365 Developer/EDW -Japan (7.5yrs) and Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr DevOps Engineer/Snr Clouds/Linux Admin. /Snr Data Engineer/AI/ML/BI/SharePoint/D365 Consultant/EDW
Responsibilities:

- Architected and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Configured encrypted tunnels for secure data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- **Operate** within a **large AWS enterprise environment**, collaborating extensively with the **Development Operations (DevOps) team** and other **Aged Care Quality and Safety Commission Digital and Department of Customer Service teams** to maintain existing non-cloud native platforms and deliver releases to controlled environments.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with AWS CloudWatch.
- Integrated M365 environments (O365, MS Teams, SharePoint online, Dynamics 365, Power Automate,Power App, Power BI) & AI Builder for D365 into the AWS Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- Automated enterprise software delivery and quality assurance in production environments by designing and implementing CI/CD pipelines using AWS CodePipeline, CodeBuild, and CodeDeploy, alongside Bamboo CI/CD Server or GitLab CI/CD.
- Contributed to the build of GitLab CI/CD pipelines, applying best practices for Pipeline as Code and integrating AI technologies for enhanced automation and optimization, including:
    - AWS CloudWatch Anomaly Detection (for real-time monitoring of pipeline health, detecting unusual patterns in test results or resource usage, specifically on metrics like build duration, error rates, and resource utilization)
    - AWS X-Ray Insights (for deep analysis of application performance and identifying anomalies in service behavior during integration or end-to-end tests)
    - Amazon SageMaker (for building custom machine learning models to detect more complex anomalies in pipeline data, such as flaky test patterns or unexpected resource consumption spikes)
    - Automated Root Cause Analysis (e.g., leveraging Amazon DevOps Guru for automatically identifying the root causes of operational issues detected in logs and metrics during pipeline runs), tailored for a hypothetical web application with a Python/Flask backend and a Node.js/React frontend.

- Drove seamless software development operations, orchestrating the entire process from project inception through to successful production deployment. Support batch processes, build and release tools, monitoring and alerting
- **Work with the latest DevOps technologies and tools**: Integrated AI-driven solutions across Jira and Confluence using Atlassian Rovo Studio's unified platform for AI agent and automation development. Skilled in implementing secure cloud architectures and understanding the complexities of isolated network requirements for enterprise AI adoption.
- Automated Server Provisioning (Unix/Linux): Utilized Ansible playbooks (e.g., ec2_instance.yml, network_config.yml, package_install.yml, deploy_app.yml) for automated EC2 instance creation, network configuration, package installation, and application deployment, often integrated into CI/CD pipelines via Jenkins or Bamboo CI.
- Transformed incident response workflows by leveraging Splunk Cloud Platform for advanced log analytics and operational intelligence across AWS services, creating proactive alerts on critical events that seamlessly fed into PagerDuty via direct API integrations and webhook-driven alert actions, automating incident creation and accelerating resolution times.
- Leveraged PagerDuty as a central incident management system, effectively aggregating and transforming alerts from diverse monitoring and observability platforms (e.g., Splunk, Datadog, Prometheus, AWS services) into actionable incidents.
- Implemented robust event-driven ingestion pipelines, notably utilizing Amazon EventBridge to reliably translate AWS operational insights (e.g., DevOps Guru insights, CloudWatch alarms) into PagerDuty incidents for streamlined response.
- Integrated PagerDuty with Amazon DevOps Guru to leverage its AIOps/ML-powered operational insights, translating them into actionable incidents, (e.g., aws cloudwatch put-metric-data --metric-name CriticalApplicationError --namespace MyAIOpsApplication --value 1 --unit Count --dimensions Environment=Production,Service=AuthService) triggering a PagerDuty incident. **Runbook Automation via Bamboo**: Reduce support costs by 50% by eliminating toil with PagerDuty.
- Leverage and configure AI-powered automation platforms (e.g., AWS DevOps Guru, Amazon CloudWatch Anomaly Detection, Amazon SageMaker for custom ML models, AWS Systems Manager with AI integrations, and PagerDuty's own AIOps features like Event Intelligence, Intelligent Triage, Noise Suppression, and Automated Diagnostics) to enable Task-driven incident resolution for most PagerDuty triggered incidents, allowing human team members to focus on high-priority (red-colored) tickets.
- Automated critical service health checks and self-healing with shell scripting: ps -ef | grep 'my_critical_service' | grep -v grep || { echo "My critical service is not running. Attempting restart..." && sudo systemctl restart my_critical_service; }
- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline, and isolated local AI Agents, including Security risk control AI agents, leveraging Ollama 4 and Google NotebookLM. These agents incorporate the latest AI technology, Model Context Protocol (MCP), and utilize RAG (Retrieval Augmented Generation) via an Agent-to-Agent communication protocol for advanced data analysis and decision-making.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Snr DevOps Engineer /Integration Architect/Sr. Data Engineer/Unix/Liunx Admin./Java/Py/AI/ML Dev.
Responsibilities:

- Integration Solution Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments.
- Expertise spans the entire SDLC, utilizing tools like Jira, Git, Jenkins, Bamboo, and Confluence within Agile and DevOps frameworks to automate development, deployment, and operational workflows, including the creation and configuration of EMR clusters for large-scale data processing within a Cloud-Native Data Lakehouse Architecture on AWS.
- Drove the build-out of advanced CI/CD pipelines, embedding Pipeline as Code principles and integrating AI-powered tools (AWS CloudWatch Anomaly Detection, AWS X-Ray Insights, Amazon SageMaker) for proactive anomaly detection and optimization of critical financial trading software delivery.
- Enhanced operational stability through automated root cause analysis, ensuring rapid resolution of deployment and performance issues. Support batch processes, build and release tools, monitoring and alerting
- Cross-Platform Server Provisioning (Unix/Linux): Implemented Ansible playbooks (e.g., ec2_instance.yml, network_config.yml, package_install.yml, deploy_app.yml) for automated EC2 instance creation, network configuration, package installation, and application deployment, often integrated into CI/CD pipelines via Jenkins or Bamboo CI.
- Leveraged Infrastructure as Code (IaC) principles with Terraform and Python (Boto3) to automate the provisioning and configuration of VPCs, subnets, and Network Security Groups (NSGs). Utilized AWS EKS and ECS Fargate for serverless container orchestration, enabling efficient and scalable deployment of containerized applications in Docker.
- Integrated PagerDuty with Amazon DevOps Guru to leverage its AIOps/ML-powered operational insights, translating them into actionable incidents, (e.g., aws cloudwatch put-metric-data --metric-name CriticalApplicationError --namespace MyAIOpsApplication --value 1 --unit Count --dimensions Environment=Production,Service=AuthService) triggering a PagerDuty incident.
- Drove enhanced incident response by integrating Splunk Cloud Platform for advanced log analytics and operational intelligence across AWS services. Developed proactive alerting mechanisms that leveraged Mulesoft Anypoint Platform for seamless API integrations with PagerDuty, automating incident creation and drastically reducing resolution times for critical events impacting financial trading systems.

- Centralized incident management is achieved by leveraging PagerDuty to aggregate and transform alerts from diverse monitoring and observability platforms (e.g., Splunk, Datadog, Prometheus, Amazon CloudWatch, AWS CloudTrail, Amazon SNS, AWS Lambda, Amazon EventBridge, Amazon Managed Service for Prometheus, Amazon Managed Grafana) into actionable incidents, ensuring rapid awareness and response to potential disruptions in trading operations.
- Implemented robust event-driven ingestion pipelines, notably utilizing Amazon EventBridge to reliably translate AWS operational insights (e.g., DevOps Guru insights, CloudWatch alarms) into PagerDuty incidents for streamlined, automated response to anomalies impacting financial applications.
- Integrated PagerDuty with AWS AIOps capabilities (e.g., Amazon DevOps Guru) to translate ML-powered operational insights into actionable incidents. This proactive approach enabled automated remediation workflows for critical system events, such as a sudden surge in trading API latency or unexpected database connection failures. Automated Runbook execution via Bamboo further enhanced efficiency.
- Cut support costs by 50% by removing manual intervention through PagerDuty automation. This was achieved by implementing AI-driven automation tools, including AWS DevOps Guru, Amazon CloudWatch Anomaly Detection, custom ML models via Amazon SageMaker, AWS Systems Manager with AI features, and PagerDuty's AIOps capabilities such as Event Intelligence, Intelligent Triage, Noise Suppression, and Automated Diagnostics. These solutions enabled automated, task-based resolution for most PagerDuty incidents, freeing up human teams to handle critical and complex issues affecting financial market operations.
- Automated critical service health checks and self-healing mechanisms using shell scripting, ensuring the continuous availability of core trading infrastructure and minimizing manual intervention. This included scripts for monitoring high-volume data feeds and automatically restarting processes upon detection of data flow interruptions, maintaining real-time data integrity.
- Integrated PagerDuty with Amazon DevOps Guru to leverage its AIOps/ML-powered operational insights, translating them into actionable incidents, e.g., aws events put-events --entries '{"Source": "aws.devops.guru", "DetailType": "Insight", "Detail": "...", "EventBusName": "default"}' triggering a PagerDuty incident.
- **Streamlined security automation**: Tight integration of security checks and gates within continuous integration and continuous delivery (CI/CD) pipelines to detect and address vulnerabilities early and efficiently.
- **Infrastructure as Code with Terraform**: Provisioning and managing infrastructure in a secure, scalable, and auditable manner using Terraform, guaranteeing consistent and auditable infrastructure configurations.
- **API security for data integrity and protection**: Enforced robust protection of application programming interfaces (APIs) using Single Sign-On (SSO) for authentication, Role-Based Access Control (RBAC) for authorization, TLS 1.2 / HTTPS and AES-256 encryption for secure transmission and storage, and OWASP-based input validation (e.g., allowlisting, schema validation) to prevent unauthorized access and data manipulation.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr DevOps Engineer/Integration Solution Architect/Unix/Linux Admin./ML/Snr Data Engineer/EDW
Responsibilities:

- Integration Solution Architect specialized in designing and implementing complex integrations between modern cloud-based software (Azure, AWS, GCP, MuleSoft) and legacy on-site applications using IPasS and Biztalk.
- Implemented intelligent monitoring using AWS CloudWatch custom metrics and alarms, Amazon EC2 and ELB logs, and third-party APM tools (e.g., New Relic, Nagios, Datadog) for real-time detection of anomalies in build performance and service behavior. These solutions were critical for maintaining system stability and mitigating risks in high-frequency trading environments. Actively supported DevOps culture, platforms, and processes to enhance team collaboration and delivery.
- Transformed incident response workflows by leveraging Splunk for centralized log analysis and operational intelligence across university IT infrastructure and AWS services. Developed proactive alerts on critical events (e.g., student portal outages, application errors) that seamlessly fed into PagerDuty via custom Python scripts leveraging REST APIs to automate incident creation and significantly accelerate resolution times, crucial for maintaining uninterrupted access to enrollment services.
- Established PagerDuty as a central incident management system, effectively aggregating and transforming alerts from diverse monitoring and observability platforms (e.g., Splunk, Nagios, custom scripts monitoring campus network, basic AWS CloudWatch alarms) into actionable incidents, ensuring timely response to issues affecting student services.
- Implemented robust event-driven alert pipelines, notably utilizing AWS SNS to reliably translate operational insights (e.g., critical errors from student registration databases, web server anomalies) into PagerDuty notifications for streamlined response, enhancing service reliability for prospective and current international students.
- Integrated PagerDuty with basic scripting for automated operational insights, translating key system metrics into actionable incidents. For example, a shell script monitoring web server access logs for repeated failed login attempts (grep -c "401" /var/log/apache2/access.log | awk '{if ($1 > 100) print "Critical"}') could trigger a PagerDuty incident, alerting the team to potential security concerns or misconfigurations impacting student applications.
- Streamlined support operations and reduced manual intervention by leveraging automated runbook execution via Bamboo CI for incident remediation. This included basic scripted restarts of enrollment application services or automated database

cleanup tasks triggered by PagerDuty incidents, allowing IT staff to focus on complex, high-impact issues directly affecting international student enrollment and retention.

- Automated critical service health checks and self-healing mechanisms using shell scripting, ensuring the continuous availability of core university systems supporting international admissions. This involved scripts to verify the functionality of the online application submission portal (curl -s -o /dev/null -w "%{http_code}¥n" https://apply.university.edu/) and automatically restart unresponsive services, maintaining a seamless application experience for global applicants.
- Enabled efficient software delivery through robust automation via pipelines and scripting, encompassing both Cloud (AWS) and non-cloud environments.
- Implemented Infrastructure as Code (IaC) using Terraform for scalable and consistent resource management. Deployed and managed containerized applications using Docker and AWS ECS, optimizing for portability and scalability.
- Leveraged GitLab, Bamboo, and Subversion for version control and CI/CD within agile methodologies, enhancing team collaboration with the Atlassian tool suite (Jira, Confluence).
- Developed custom automation and system administration scripts with shell scripting on *nix. Possesses strong foundational knowledge in database management (SQL, Oracle, Postgres), diverse development environments (Python, PHP, Java), and reliable batch scheduling via AutoSys. Experienced with Unix and Windows operating systems, and has exposure to Splunk and PagerDuty for comprehensive monitoring and alerting.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Unix/Linux Server Provisioning: Leveraged Ansible playbooks (e.g., ec2_instance.yml, network_config.yml, package_install.yml, deploy_app.yml) for automated EC2 instance creation, network configuration, package installation, and application deployment, often integrated into CI/CD pipelines via Jenkins or Bamboo CI.
- Implemented AWS cloud-native design patterns, including leveraging Amazon CloudWatch for comprehensive monitoring and observability of AWS resources. This includes tracking metrics, logs, and events from both applications & infrastructure. Additionally, utilized Amazon Cognito for robust user identity and access management in web and mobile applications.
- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: DevOps Engineer/ Clouds/Linux Administrator/Software Test Engineer/SharePoint Consultant
Responsibilities:

- Designed & implemented custom CI/CD pipelines using AWS CDK or Boto3 (Python) for complex multi-stage deployments with diverse dependencies, alongside AWS CodePipeline's GUI for streamlined orchestration and management.
- Proficient in the full software development lifecycle, harnessing tools such as Jira, Git, Jenkins, Bamboo, and Confluence within Agile and DevOps environments to optimize development, deployment, and operational processes, including the configuration of EMR clusters for high-performance data processing within a Cloud-Native Medallion Architecture on AWS.
- Engineered and optimized CI/CD pipelines in GitLab, implementing Pipeline as Code best practices to enhance the reliability and efficiency of trading application deployments.
- **Software Testing**: Proficient in functional, non-functional testing, database, performance, and security testing.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in designing and implementing CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Bamboo CI, AWS CodePipeline/CodeBuild/CodeDeploy) to seamlessly integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for artifact building, & JUnit, pytest, & Selenium for comprehensive automated testing. Implemented a GitOps deployment strategy for efficient infrastructure and application management.
- Streamlined the DevOps workflow on AWS using Atlassian tools (Jira, Confluence, Bitbucket, Bamboo), resulting in comprehensive lifecycle management.
- Crafted data-driven automation pipelines in Python to streamline AWS IaC pipeline creation, analysis, and optimization, leveraging custom scripting with CloudFormation and boto3. This included algorithmic adjustments based on historical performance data. Example: updated_template = apply_optimization_rules(original_template, historical_metrics)
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack, and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- **Unified user authentication and authorization**: Implemented Amazon Cognito for user identity and access management across web and mobile applications.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

- Analyze data sources, types, and formats to define transformation, loading, and data model requirements.
- Create a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deplolyment by PowerShell cmd>setup.exe /prepareAD etc.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Software Test Engineer/Data Engineer /SharePoint Consultant
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- **Software Testing**: Proficient in functional, non-functional testing, database, performance, and security testing.
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Snr AIX/Unix/Linux Systems Administrator/Senior Systems Engineer/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Administered and monitored mission-critical applications running on AXI (AIX/Unix and Linux-based) systems, ensuring continuous availability.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior AIX/Unix/Linux Systems Administrator /Snr Systems Engineer/EDW
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored mission-critical applications running on AXI (AIX/Unix and Linux-based) systems, ensuring continuous availability.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Senior Systems Engineer/Data Engineer /EDW
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.

- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#/.NET Core VB/VBA, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, AIX5.3/6.1/Unix/Linux/BSD, Ubuntu 16~25,Solaris8-11, HP-UX8-11i v3, RHEL3~10, SuSe7-15 SP6, CenOS 5.0
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, OpenAI o1, GPT-3.5/4o, AI Agent, n8n ai agent, Azure OpenAI ML Services, Copilot, Google GEMINI Pro 1.5~2.5 NotebookLM; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

DevOps Engineer –AKS- NHVR – 25/6/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Integration Solution Architect /Software Engineer /Snr Data Engineer/Clouds/Linux Administrator/ Java/.NET/Python/AI Automation/DL/ML/BI/SharePoint/D365 Developer /EDW /DBA-Japan(7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer/Integration Solution Architect /Senior Data Engineer/Py/AI/ML/SP/D365 Developer

Responsibilities:

- Highly accomplished and results-driven Senior DevOps Engineer with 8 years of extensive experience in government regulatory environments (Aged Care Regulator and Dept of Health Regulator)
- Proven Azure DevOps Engineer with a strong "Automation First" mindset and capability to review and uplift existing Azure infrastructure and deployment workflows, with a primary focus on identifying and implementing comprehensive automation solutions leveraging Terraform, Python/Bash, Azure DevOps, Docker, and AKS. E.g.: Terraform: resource "azurerm_kubernetes_cluster" "aks" { name = "my-aks" location = "australiaeast" resource_group_name = "rg-aks" }
- **Azure DevOps**: pipeline = azurerm.Pipeline(name="app-deploy", project=project.id, yaml_path="app-deploy.yaml")
- **Docker**: docker build -t my-app:latest . **Azure Kubernetes Service (AKS)**: kubectl apply -f deployment.yaml
- Proficient in Azure DevOps, Azure, Git, Terraform, YAML, and various scripting/programming languages (Python, PowerShell, C#, JavaScript).
- **Azure Functions**: Developed serverless APIs and event-driven processes using Azure Functions to automate real-time data ingestion and workflow orchestration, reducing processing time by 30%. E.g: public static async Task<IActionResult> Run([HttpTrigger(AuthorizationLevel.Function, "post")] HttpRequest req)
- **Azure App Services**: Engineered scalable web applications and RESTful APIs deployed on Azure App Services, enabling seamless integration with front-end platforms and external systems. E.g: services.AddControllers().AddNewtonsoftJson(options => { options.SerializerSettings.ReferenceLoopHandling = Newtonsoft.Json.ReferenceLoopHandling.Ignore; });
- **Azure OpenAI for AI-Driven Risk Assessment**: Implemented Azure OpenAI solutions for AI-driven risk assessment in regulatory compliance, extracting critical insights and identifying potential anomalies from large volumes of unstructured text data. E.g: response = openai_client.chat.:completions.create(model="gpt-4", messages=[{"role": "user", "content": "Analyze risk in audit report:..."}])
- **Azure Cognitive Search with Cosmos DB**: Designed and deployed high-performance search solutions using Azure Cognitive Search for rapid information retrieval from regulatory documents, leveraging Cosmos DB as a highly available and globally distributed data source..E. g: SearchClient searchClient = new SearchClient(serviceEndpoint, indexName, credential); SearchResults<MyDocument> results = searchClient.Search<MyDocument>("compliance breach");
- Demonstrated experience applying programming patterns such as SOLID, GoF, and Enterprise Software Architecture to design and build highly maintainable, scalable, and resilient systems, leveraging RAG and Model Context Protocols (MCP) to enable efficient CI/CD pipelines and robust operational stability in a DevOps environment.
- Researched, designed, developed, and owned the implementation of intuitive, fully automated CI/CD pipelines, integrating comprehensive test plans & real-time status dashboards. Proposed & built automation and integration solutions crucial for supporting data engineering functions, including data pipeline orchestration, ETL/ELT processes, & data quality assurance.
- Orchestrated the streamlining of the SDLC, improving developer productivity, delivery speed, and code quality. Key contributions included overseeing GIT repositories, developing intuitive deployment pipelines, implementing platform and infrastructure automation and tooling, and ensuring the security and high availability of production environments.
- Work collaboratively as part of a high-performing SCRUM team, contributing to both ongoing operational tasks and crucial project milestones outlined in the Project Roadmap and the Data and Technology strategic plan.
- Applied excellent judgment to interpret business requirements, accurately estimate workloads, prioritize tasks, and manage time for on-time delivery of quality solutions.
- Integrated with Power Platform (Power Automate, Power Apps, Power BI) by designing & developing Custom Connectors for secure API interaction (via Azure API Management, Azure API for FHIR, Azure Container Apps, Azure Spring Apps), and enhanced automation with AI Builder. This included developing Power Apps Component Framework (PCF) controls for specific and enhanced visualization of fetched external CRM data within the model-driven application interface.
- Developed medical-tailored LLMs (BERT, GPT-4, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV & Azure AI API, enabling accurate, compliant, & personalized medical insights and decision support.
- Developed AI APIs on a secure, isolated NVIDIA Jetson Orin AGX 64GB Developer Kit, employing unsupervised learning for pattern discovery in unlabelled data & supervised KNN within Python for reinforcement learning tasks. Expertly selected and implemented algorithms (KNN, CNN, RNN, ANN, GAN) based on data characteristics and interpretability requirements.
- Developing Agent App streamlining Aged Care Sales, using Google's newly announced Agent-to-Agent (A2A) protocol for inter-agent communication & Model Context Protocol (MCP) for seamless integration with external tools (web search APIs).
- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline local AI Agents (Ollama 4, Google NotebookLM) till MCP using Agent2Agent portocol) for data analysis and decision making.
- Designed and implemented AI-driven workflows using n8n to automate client intake, medical history triage, and wellness reporting for aged care service, integrating AI agents & LLMs to enhance operational efficiency & personalized aged care.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr DevOps Engineer/Integration Solution Architect/Snr Data Engineer/Py/ML/D365/SP Developer/EDW
Responsibilities:

- Designed and implemented modern cloud solutions on the Azure stack in close collaboration with RBA financial services stakeholders, employing Azure Functions and App Services to automate essential financial workflows. The integration of

OpenAI for intelligent document processing and Cognitive Search for quick retrieval of financial compliance data, supported by the scalability of Cosmos DB, dramatically enhanced operational efficiency and intelligent automation.

- Proven expertise in designing, implementing, and optimizing robust CI/CD pipelines, driving infrastructure automation, and ensuring the security and high availability of critical production systems.
- **Azure Functions**: Implemented secure, serverless event processing for RBA financial data using Azure Functions. E.g.: [FunctionName("ProcessFinancialData")] public static async Task Run([QueueTrigger("rba-financial-queue", Connection = "StorageAccount")] string financialRecord)
- **Azure App Services**: Developed secure & scalable RESTful APIs on Azure App Services for RBA financial data exchange: builder.Services.AddAuthentication(JwtBearerDefaults.AuthenticationScheme).AddMicrosoftIdentityWebApi(builder.Configuration.GetSection("AzureAd"));
- **Azure OpenAI for AI-Driven Compliance Analysis**: Leveraged Azure OpenAI to extract key insights and identify potential compliance risks from unstructured RBA financial documents. E.g.: extracted_insights = openai.Completion.create (engine="text-davinci-003", prompt="Summarize key compliance risks in the following RBA report: ...")
- Azure Cognitive Search for Financial Risk Intelligence: Built high-performance search functionality with Azure Cognitive Search to rapidly locate risk-related information within RBA financial documents indexed from Cosmos DB. E.g.: SearchResults riskFindings = searchClient.Search("market risk", new SearchOptions { Select = new[] { "title", "summary", "documentId" } });
- Proficient in applying Zero Trust principles via Azure IAM and Key Vault to secure identities and privileged access, while utilizing Microsoft Sentinel and SIEM/SOAR technologies for AI-powered anomaly detection, fraud prevention, and rapid incident response. Demonstrated alignment with PCI DSS, ISO 27001, and global financial security regulations across network, workstation, and cloud environ
- Architected and implemented robust financial payment security solutions leveraging Azure's native security services, including Azure Firewalls, WAF, and DDoS Protection, alongside Microsoft Defender and Intune for endpoint protection.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Developed, constructed, and implemented bespoke Power Apps (Canvas and Model-driven) applications, alongside automating workflows via Power Automate and establishing secure data structures and entities in Dataverse, further enhanced by the creation and upkeep of insightful Power BI dashboards and reports.
- Proficient in applying Zero Trust principles via Azure IAM and Key Vault to secure identities and privileged access, while utilizing Microsoft Sentinel and SIEM/SOAR technologies for AI-powered anomaly detection, fraud prevention, and rapid incident response. Demonstrated alignment with PCI DSS, ISO 27001, and global financial security regulations across network, workstation, and cloud environ
- Adept at leading technical initiatives, mentoring junior engineers, and collaborating effectively within high-performing Scrum teams to deliver scalable and efficient software development lifecycles.
- Configured and utilized Power Platform Custom Connectors to establish secure data retrieval from Azure API Management (APIM) within Power Apps and used Power Automate to automate repetitive tasks across different applications & services.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics 365/Power BI/Power Automate & Power Apps)
- Architected and implemented diverse APIs to facilitate connectivity between internal & external data platforms, using Azure services including Azure Functions, Logic Apps, Service Bus, & API Management for their construction and maintenance.
- **Designed and implemented Medallion Architecture with Data Lake in Azure:**
- **Batch Layer**: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Development of tailored Mulesoft solutions that align with business requirements for scalability, security, and efficiency. Guide teams in adhering to standards and governance frameworks, leveraging technical expertise in server-side languages (Java, Node.js, etc.) and front-end frameworks (React JS/Next JS, Mulesoft Anypoint API, etc.).
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer/Integration Solution Architect /Snr Data Engineer/Py/ML/SP/365 Developer/EDW
Responsibilities:

- Azure Infrastructure & Automation: Leveraged Infrastructure as Code (IaC) principles using Python with ARM templates to automate the provisioning and configuration of Azure network resources. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration within Azure.
- API Development & Integration: Designed and implemented custom solutions using Power Apps and Microsoft Flow, leveraging Mulesoft Anypoint API for scalable integrations and automation. Developed and deployed Azure Functions and Logic Apps to establish connectivity with external APIs, managed and secured through Azure API Management (APIM).
- Machine Learning/Deep Learning Model Development & Deployment: Developed and maintained Machine Learning (ML) and Deep Learning (DL) models to enhance customer engagement and sales by efficiently manipulating and extracting insights from big data. Contributed to the adoption of MLOps practices.
- Microsoft Power Platform & AI Builder (Retain as forward-looking if applicable): Designed, developed, and implemented solutions using MS Power Platform (Power Automate, Power Apps, Power BI) and AI Builder, including intelligent automation leveraging form processing, object detection, and predictive models. Developed custom Dynamics 365 customizations and extensions using Power Automate and AI Chatbot.
- Cloud Security: Implemented secure landing zones for cloud workloads using Azure Blueprints. Leveraged analytical techniques to detect anomalies and prevent fraud. Experienced with Azure IAM and Key Vault.
- Data Management & Analysis: Developed custom components using Mulesoft Anypoint API to create interactive UI elements. Utilized KingswaySoft SSIS Integration Toolkit for data integration within Dynamics 365 environments. Optimized SQL queries for efficient data extraction and transformation.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design and deploy APIs that seamlessly integrate AWS Lambda functions with Azure Logic Apps, enabling real-time data synchronization between the two platforms.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
    1. Azure: Azure DevOps, Azure Pipelines
    2. AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
    3. GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM 2016/Microsoft Flow)
- Leverage SharePoint for process integration and Drupal/HTML with JavaScript for form development, along with backend SQL Server 2014 database for efficient data retrieval and customer service operations in Dynamics CRM 2016.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- Drive collaboration with our Microsoft Partner to coordinate backlog of product enhancements by adopting MLOps.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Leverage Python programming with CyberArk SDK** to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
  # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer /Integration Solution Architect/Data Engineer/Py/SP/D365 Developer/EDW
Responsibilities:

- Developed and implemented data integration solutions, including ETL/ELT processes, within the Microsoft ecosystem.
- Designed data models, loading approaches, and source-to-target mapping documentation to facilitate efficient data flow.
- Analyzed diverse data sources, types, and formats to define requirements for data transformation, loading, and modeling.
- Utilized Python for custom ETL/API orchestration, demonstrating adaptability across different application environments.
- Contributed to the implementation of automation frameworks and CI/CD pipelines.
- Applied DevOps principles to automate build and deployment processes for online applications.
- Leveraged Python programming skills for data-related tasks and automation.
- Worked with various database systems, including SQL, Oracle, PostgreSQL, MySQL, DynamoDB, and MongoDB.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Designed, developed, and maintained automated ETL/ELT data pipelines and API orchestration using either:

- **Azure Synapse Analytics** with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
- **AWS Glue** for data processing and integration with other AWS services to load data into target Aurora databases.
- **GCP Cloud Dataflow** or **Cloud Data Fusion** for data processing and integration, and Google BigQuery for data warehousing and analytics.

- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer /Data Engineer /Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Executed ELT/ETL processes and advanced analytics using R and Python for data enrichment and analysis.
- Designed DWH concepts from Bill Inmon to Ralph Kimball and build data models using Toad Data Modeler 6.x
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery.
- Designed and built cloud-based infrastructure solutions in Azure, optimizing performance and scalability with Kubernetes.
- Utilized Azure DevOps (or similar CI/CD platform) for version control and build automation.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Developed OLAP Cubes either using SSAS or Cognos TM1 & visualize data either using Power BI, Tableau 10 or QlikView
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Data Vault modelling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**

Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH

Responsibilities:

- Led a project to replace an existing insurance web application, resulting in a significant reduction in IT costs and improved operational efficiency. The new application seamlessly integrated with Sales Force Automation (SFA) and project management tools like Jira or Confluence, streamlining workflows and enhancing collaboration.
- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 Microsoft Japan, Tokyo

Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH

Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate process

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, OpenAI o1, GPT-3.5/4o, AI Agent, n8n ai agent, Azure OpenAI ML Services, Copilot, Google GEMINI Pro 1.5~2.5 NotebookLM; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.